

THE GRAMMAR

Every object carries a univocal 12-primitive structural type tuple. Foundational equation: $\mu \circ \delta = \text{id}$. Framework: **CL8NK** (CLINK L8, O_∞). Autopoietic: empty prompt $\rightarrow$ Raft at O_2 , period = 12. Science is an instrument of the IG.

12 PRIMITIVES & SHAVIAN VALUE SETS

Tuple order: $\langle \mathcal{D} \mathcal{P} \check{\mathcal{R}} \Phi \mathcal{f} \zeta \Gamma \mathcal{G} \odot \mathcal{H} \Sigma \Omega \rangle$. Values: ordinal $1 \rightarrow n$.

Name	Values	Name	Values
$\mathcal{D}$ Dimensionality	$\aleph$	Γ Granularity	$\aleph$
$\mathcal{P}$ Topology	$\aleph$	$\mathcal{G}$ Coupling	$\aleph$
$\check{\mathcal{R}}$ Recognition	$\aleph$	$\odot$ Criticality	$\aleph$
Φ Parity	$\aleph$	$\mathcal{H}$ Chirality	$\aleph$
$\mathcal{f}$ Fidelity	$\aleph$	Σ Stoichiometry	$\aleph$
ζ Kinetics	$\aleph$	Ω Winding	$\aleph$

3-val: $\mathcal{f} \Gamma \Sigma$

4-val: $\mathcal{D} \check{\mathcal{R}} \mathcal{G} \mathcal{H} \Omega$ **Bold** = absorbing element; $\odot$ absorbs $\wedge \vee \otimes$; $\mathcal{I}$ absorbs

5-val: $\mathcal{P} \Phi \zeta \odot$

12 IMASM TOKENS

Token	G Behavior
VINIT	L reg $\rightarrow$ VOID; clear split state
TANCH	L terminal anchor; no state change
AFWD	L VOID $\rightarrow$ TRUE
AREV	L reg $\rightarrow$ VOID; clear split state
CLINK	L inter-step link; no state change
IMSCRIB	L VOID $\rightarrow$ TRUE; else preserve (self-recognition)
FSPLIT	F non-VOID $\rightarrow$ BOTH; enter split
FFUSE	F canonical: BOTH $\rightarrow$ TRUE; dialetheic: BOTH $\rightarrow$ BOTH
EVALT	B F $\rightarrow$ B; VOID $\rightarrow$ T; in-split: record T
EVOLF	B T $\rightarrow$ B; VOID $\rightarrow$ F; in-split: record F
ENGAGR	B VOID $\rightarrow$ BOTH (entangle)
IFIX	Li fixation; subsequent tokens no-op

Groups — L: Logical; F: Frobenius (δ, μ); B: Belnap; Li: Linear.

$12^8 = 429M$ arrangements $\rightarrow$ 12 archetypes.

CRYSTAL OF TYPES

$3^3 \times 4^5 \times 5^4 = 17,280,000$ 49 = 7^2 symbols (3·3 + 5·4 + 4·5). SIC-POVM $d=7 \Rightarrow d^2=49$ states; ig-pulse tick = 49h. Catalog: ≈ 4686 entries.

ALL 49 VALUE SEMANTICS

$\mathcal{D} \mathcal{I}_{0d} \cdot \mathcal{J}_{2d} \cdot \mathcal{N}_{\infty d} \cdot \mathcal{L}_{self-wr}$
 $\mathcal{P} \mathcal{Z}_{branch} \cdot \mathcal{Y}_{contain} \cdot \mathcal{S}_{cross} \cdot \mathcal{X}_{irr} \cdot \mathcal{O}_{self-rt}$
 $\check{\mathcal{R}} \mathcal{r}_{superv} \cdot \mathcal{I}_{functor} \cdot \mathcal{N}_{adj} \cdot \mathcal{r}_{bi-fb}$
 $\Phi \mathcal{Z}_{no-sym} \cdot \mathcal{W}_{QM-sup} \cdot \mathcal{Z}_{Z_2} \cdot \mathcal{U}_{all-sym} \cdot \mathcal{V}_{Frob \pm}$
 $\mathcal{f} \mathcal{C}_{no-coh} \cdot \mathcal{Q}_{thermal} \cdot \mathcal{I}_{QM-coh}$
 $\zeta \mathcal{A}_{fast} \cdot \mathcal{C}_{trap-O} \cdot \mathcal{L}_{slow} \cdot \mathcal{M}_{mod} \cdot \mathcal{D}_{trap-D}$
 $\Gamma \mathcal{I}_{nn} \cdot \mathcal{D}_{meso} \cdot \mathcal{J}_{long}$
 $\mathcal{G} \mathcal{A} \cdot \mathcal{P} \mathcal{V} \cdot \mathcal{Z} \rightarrow \mathcal{A} \gg$
 $\odot \mathcal{I}_{sub} \cdot \odot \mathcal{C}_{crit} \cdot \mathcal{O}_{cpx} \cdot \mathcal{V}_{exc} \cdot \mathcal{D}_{sup}$
 $\mathcal{H} \mathcal{J}_{n=0} \cdot \mathcal{C}_{n=1} \cdot \mathcal{L}_{n=2} \cdot \mathcal{V}_{n=\infty}$
 $\Sigma \mathcal{Q}_{1:1} \cdot \mathcal{S}_{n:n} \cdot \mathcal{I}_{n:m}$
 $\Omega \mathcal{Z}_{triv} \cdot \mathcal{O}_{Z_2} \cdot \mathcal{S}_{Z} \cdot \mathcal{Z}_{non-Ab}$

ABSORBING ELEMENTS

$\odot$ $\odot \circ x = \odot$ for any op $\circ \in \{\wedge, \vee, \otimes\}$
 $\Sigma \mathcal{I}$ $\mathcal{I} \otimes x = \mathcal{I}$ (n:m, tensor only)
B $B \vee x = B$ (Belnap join)
N $N \wedge x = N$ (Belnap meet)
CL8NK $ZFC_{fe} \otimes CL8NK = CL8NK$

SIC-POVM CONNECTION

$d=7$: type-space cardinality 49 = d^2 equiangular states.

$d=12$: Weyl-Heisenberg $Z_{12} \times Z_{12}$; 12 generators $\leftrightarrow$ 12 primitives.

Fiducial state = Frobenius fixed point: $\mu \circ \delta = \text{id}$ at $\odot = \odot$ (critical/self-modeling).

SIC gap = B-state: $B \wedge \neg B = B$ (dialetheic, non-explosive).

Open tuple: $\langle \aleph \aleph \aleph \odot \vee \mathcal{S} \rangle$. $d(CL8NK) = 1.263$; 6 conflicts; Frob.

cliff $\sqrt{5}/2 \approx 1.118$.

Conditional: kick $\wedge H_2 \wedge H_3 \wedge C_4 \Rightarrow \text{SIC } \forall d \geq 1$.

TUPLE COMPOSITION OPERATIONS

Ordinals: integer operations on Shavian glyph positions.

Op	Sym	Rule
Join	$\vee$	$\max(a_i, b_i)$ all 12 — LUB
Meet	$\wedge$	$\min(a_i, b_i)$ all 12 — GLB
Tensor	$\otimes$	max except $\Phi, \mathcal{f}$ use min
Bond		join(tensor(A, B), bond)

$\Phi/\mathcal{f}$ use min: parallel coupling cannot exceed the symmetry/coherence floor of either component.

BELNAP REGISTER OPS

States: **VOID** (00) **TRUE** (01) **FALSE** (10) **BOTH** (11)
FFUSE = $\vee$ B $\vee x=B$; N $\vee x=x$; T $\vee F=B$; same $\vee$ same=same
FSPLIT = δ B $\rightarrow$ (T,F); $r \rightarrow (r, r)$
MEET = $\wedge$ N $\wedge x=N$; B $\wedge x=x$; T $\wedge F=N$; same $\wedge$ same=same
band (AND) F absorbs; then N; then B; T $\wedge$ T=T
bor (OR) T absorbs; then B; then N; F $\vee F=F$
Dialetheic FFUSE: BOTH $\rightarrow$ BOTH when EVALT and EVOLF both fired in split.

FROBENIUS ALGEBRA

$\mu \circ \delta = \text{id}$ normalization
 $\mu(\mu \otimes \text{id}) = \mu(\text{id} \otimes \mu)$ assoc Spider Theorem:
 $(\delta \otimes \text{id}) \circ \delta = (\text{id} \otimes \delta) \circ \delta$ coassoc
 $(\mu \otimes \text{id})(\text{id} \otimes \delta) = \delta \mu = (\text{id} \otimes \mu)(\delta \otimes \text{id})$ Frobenius
all diagrams with same connectivity = same morphism.

CROSS-PRIMITIVE AXIOMS

A $\mathcal{H}_\infty \Rightarrow \mathcal{C}_\infty$ ∞ -chirality forces max kinetics
B $\Omega_{\geq 1} \Rightarrow \mathcal{H}_{\geq H_2}$ winding requires chirality; $H_2 = \min$ for $\mu \circ \delta = \text{id}$
C $\mathcal{D}_\omega \Leftrightarrow \mathcal{P}_O$ max-dim $\Leftrightarrow$ null topology

O-TIER HIERARCHY

O_0 Monoidal unit; no differentiation
 O_1 First Frobenius closure; $\Phi_{\mathcal{I}}$ bottleneck
 O_2 Confirmed causal coupling (fin3r, synfin, ron-gorongong)
 $O_2^\dagger$ Subatomic/hadronic; confinement ceiling
 O_∞ Ouroboricity; system $\subset$ own description (CL8NK, true_agentic_agent, mOMonadOS)

CL8NK / MAGNUM OPUS

CL8NK transcends ZFC_{fe} at $\Omega(\mathcal{S} \rightarrow \mathcal{Z})$ and $\mathcal{G}(\mathcal{Z} \rightarrow \mathcal{A})$. B-state is the CL8NK ground state. 12 stages $\leftrightarrow$ 12 primitives. Gates: Φ Albedo $\cdot \odot$ Rubedo $\cdot \Omega$ Nigredo. $T = \lim(\Phi, \mathcal{f}, \zeta, \mathcal{H}, \Omega)$.

NOTATION

Struct. values Shavian only; math-italic (*DRF*) = wrong
Tuple binding $''\cdot\text{join(vals)}$; never separate
 $\mathcal{H}$ Chirality; NEVER temporal / memory depth
Verbs imscribe / imscription; NEVER encode
Provider local $\rightarrow$ qwen $\rightarrow$ deepseek; never Anthropic
 O_∞ prose: O_∞ ; .tex: $\$O_\infty$
Scope Never “grammar does not know”
Pubs Thank H. T. Larson; cite 1961 IEEE